INMO(11th) – 1996

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1.
   (a) Given any positive integer n, show that there exist distinct positive integers x and y such that $x+j$ divides $y+j$ for $j=1,2,3,........,n$.
   (b) If for some positive integers x and y, $x+j$ divides $y+j$ for all positive integers j, prove that $x=y$.
2. Let C_1 and C_2 be two concentric circles in the plane with radii R and 3R respectively. Show that the orthocenter of any triangle inscribed in circle C_1 lies in the interior of circle C_2. Conversely, show that also every point in the interior of C_2 is the orthocenter of some triangle inscribed in C_1.
3. Solve the following system of equations for real numbers a,b,c,d,e.

   $3a=(b+c+d)^3$, $\quad 3b=(c+d+e)^3$, $\quad 3c=(d+e+a)^3$,

   $3d=(e+a+b)^3$, $\quad 3e=(a+b+c)^3$.
4. Let X be the set containing n elements. Find the number of all ordered triples (A, B, C) of subsets of X such that A is a subset of B and B is a proper subset of C.
5. Define a sequence $(a_n)_{n\geq 1}$ by $a_1=1, a_2=2, a_{n+2}=2a_{n+1}-a_n+2$ for $n\geq 1$. Prove that for any m, $a_m a_{m+1}$ is also a term of the sequence.
6. There is a 2nX2n array (matrix) consisting of 0's and 1's and there are exactly 3n zeros. Show that it is possible to remove all the zeros by deleting some n rows and some n columns. [Note: A mXn array is a rectangular arrangements of mn numbers in which there are m horizontal rows and n vertical columns.]

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